

> ROB HAAG

SOFTWARE · SYSTEMS · INTEGRATION ENGINEER

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PROFILE

Software and systems engineer whose work has crossed application development, IoT/RFID, Linux, AWS, networking, embedded systems, and hardware/software integration. Strongest in unfamiliar systems where the failure is not obvious and the answer has to be found through measurement, experimentation, documentation, and systematic debugging.

CORE TECHNOLOGIES

Software: C/C++, Python, Go, JavaScript / Node.js, PHP, C#, Bash, REST, WebSockets, SQL, Git/GitHub

IoT / Embedded: RFID, MQTT, BLE, ZPL, RP2040/RP2350, ESP32, GPS/GNSS, PPS, UART, SPI, I2C, RS-232

Cloud / Systems: AWS, AWS CLI, CloudFormation, Docker, Kubernetes, ECS, Lambda, Jenkins, CI/CD, Linux, TCP/IP, DNS, NTP, Nginx, HAProxy

Data / Diagnostics: PostgreSQL, MySQL, MongoDB, JSON, YAML, XML, ETL; schematics, oscilloscopes, logic analyzers, serial debugging, hardware bring-up

EXPERIENCE

Electrical / Mechanical Troubleshooting & Repair | Central Ohio | 2021-Present

Hands-on diagnostic work on vehicles and heavy equipment, with an emphasis on difficult electrical and electronic faults.

- Diagnose wiring, sensors, actuators, control modules, and electrical/mechanical interactions using schematics, direct measurement, functional testing, and systematic isolation.
- Take on unusual problems that extend beyond routine parts replacement, often with incomplete documentation and uncertain failure modes.
- Continue active software, Linux, networking, embedded-systems, and electronics development through independent engineering projects.

DevOps Engineer | Bibliotheca LLC | 2020-2021

One of two engineers responsible for the AWS/DevOps side of a year-long platform refactor and rearchitecture.

- Built and maintained AWS infrastructure, networking, load balancing, access controls, development/test environments, redundancy, recovery, and failover.
- Automated infrastructure and deployments with CloudFormation, YAML/JSON, AWS CLI, CI/CD pipelines, and supporting database environment creation/migration.
- Supported Docker, Kubernetes/ECS, Lambda, Jenkins, monitoring, and logging across the replacement platform.
- Engineered resilient infrastructure in which failed or deliberately removed components could be recreated within minutes while redundant systems maintained service.

Senior Engineer - IoT / RFID | Smartrac | 2018-2020

R&D; and systems integration spanning RFID hardware, connected devices, Linux services, telemetry, and backend software.

- Developed and integrated commercial IoT/RFID systems using MQTT telemetry, Linux, PostgreSQL, REST services, and physical RFID devices.
- Brought up, evaluated, configured, and troubleshoot unfamiliar RFID hardware and development platforms to determine actual capabilities and project fit.
- Developed ZPL workflows that encoded RFID tags during label printing; also worked with temperature-sensing RFID tags and BLE devices for cold-chain monitoring.
- Investigated failures across hardware, communications, software, databases, and infrastructure to characterize system behavior and limitations.

Programmer / Analyst | Command Alkon Inc. | 2015-2018

Production SaaS development for ready-mix concrete supply-chain operations.

- Developed and supported a Node.js / AngularJS / MongoDB application, working across frontend, backend, database, APIs, deployment, and infrastructure.
- Built REST APIs, WebSocket real-time interfaces, and external JSON/XML integrations; performed extensive bug hunting and root-cause analysis.
- Worked in a weekly production release cycle with Nginx, HAProxy, AWS EC2/S3/Lambda, Docker, JavaScript/Node.js, and Go.
- Helped coworkers during the team's Git/GitHub adoption, drawing on previous production Git experience.

Web Developer | Maxximaging | 2010-2014

Major rearchitecture of a legacy HIPAA-regulated application and its production infrastructure.

- Took over a legacy PHP application and substantially refactored/rearchitected it using Zend Framework and a maintainable layered design.
- Migrated the data-access layer to a different ORM and used Pentaho/Kettle ETL for data transformation and migration work.
- Remediated significant inherited security problems involving credential handling, input processing, authentication/access control, encryption, and sensitive-data protection.
- Moved the system from hospital-hosted infrastructure to geographically redundant AWS infrastructure and managed Linux, Apache, PHP, MySQL, Git, deployment, and production.

Front-End Developer - Contract | Abercrombie & Fitch | 2015

- Developed customer-facing fixes and features across major retail websites using HTML, CSS/SCSS, JavaScript, jQuery, JSTL/JSP, REST/AJAX, Git, Grunt, NPM, Jira, and Linux.

Freelance & Contract Software Development | 2009-2014

- Delivered full-stack web and systems work across healthcare, e-commerce, and enterprise projects using PHP, JavaScript, MySQL, Linux/Apache, AWS, Git, REST, XML/JSON, ASP.NET MVC, Ruby on Rails, and Three.js.

SELECTED CURRENT ENGINEERING PROJECTS

GPS-Disciplined Embedded Time System - RP2350/Pico 2 firmware integrating u-blox GPS/GNSS, hardware PPS, serial interfaces, and an RS-232 VFD display; includes timing qualification, hardware abstraction, and experimental characterization of incompletely documented hardware.

Embedded Network Time Appliance - ESP32 network interface for an authoritative GPS-disciplined source, including MCU-to-MCU communication, LAN NTP, Wi-Fi provisioning/status, source-health reporting, and controlled fallback.

Electronic Parts Inventory System - Practical inventory application for electronic-parts stock with application backend, persistent storage, and local deployment.

EDUCATION

Franklin University - Coursework, 2008 · Central New Mexico Community College - Coursework, 2003-2004